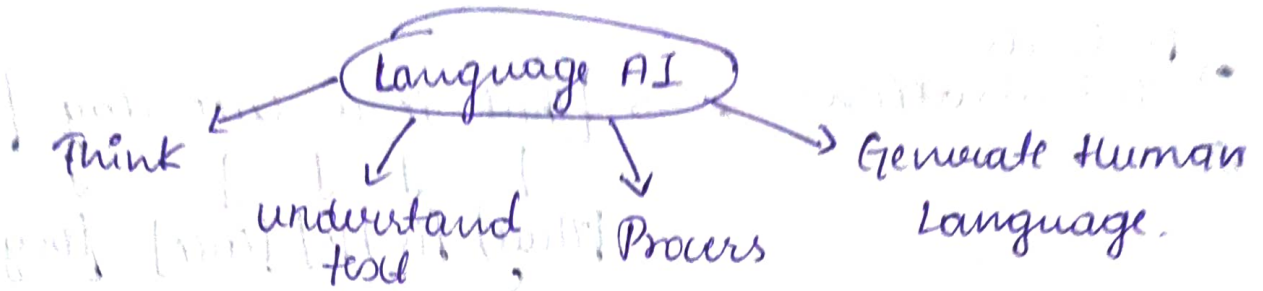


Large language model

Day 1:-

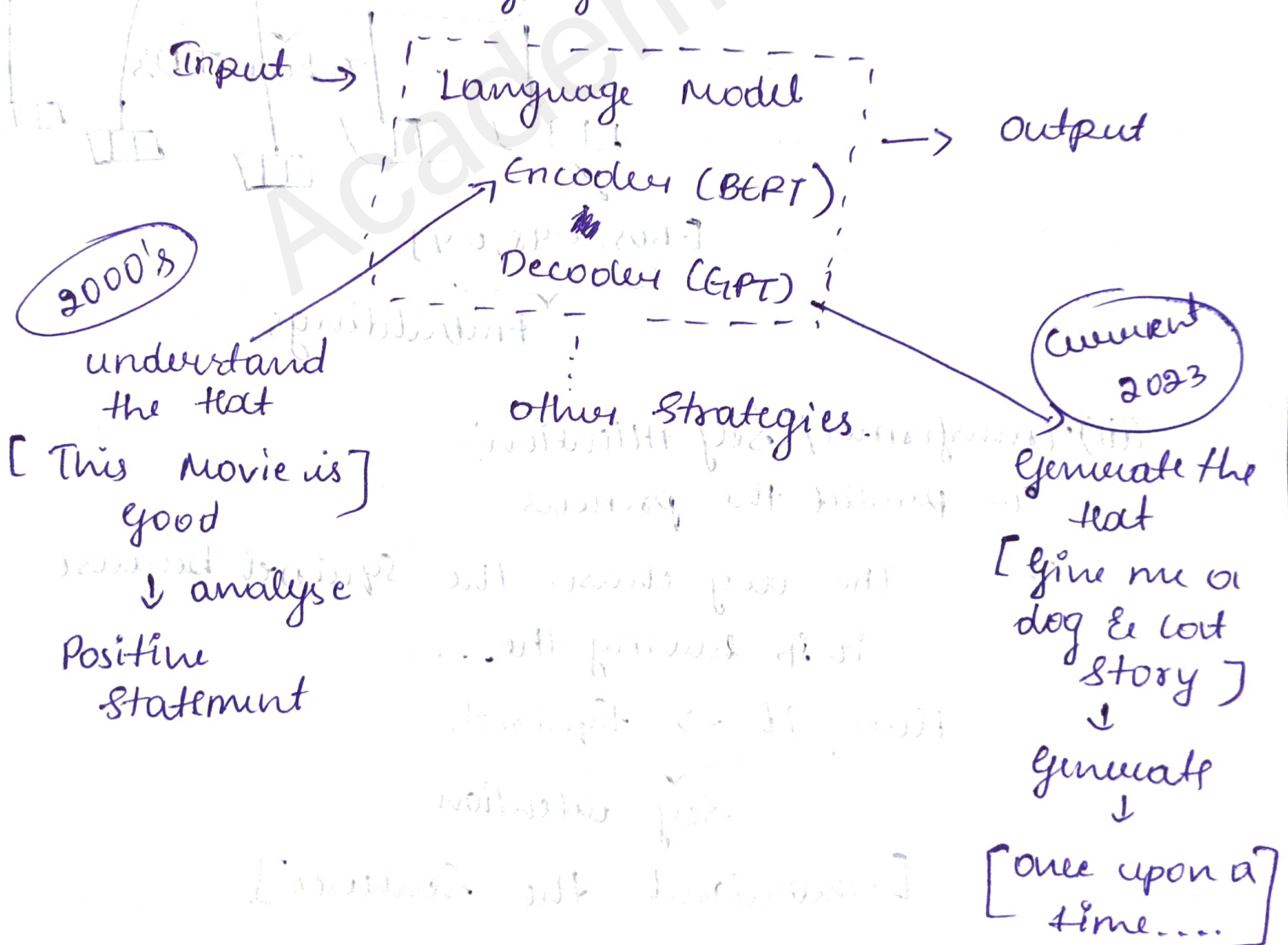
Introduction to Language AI :



This language AI uses GPT (Decoder) & BERT (Encoder)

GPT → Generate Human like language

BERT → Analyze, classify & extract Meaning from Text

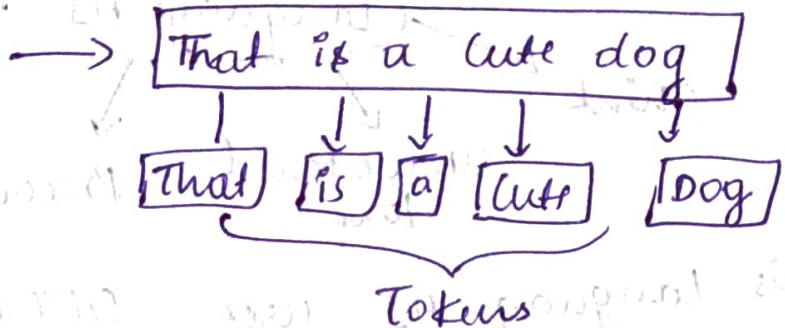


How Does Language Works:-

Computer doesn't understand human language
So it understands 0's & 1's So we need to
transform Humanised (unstructured) to structured.

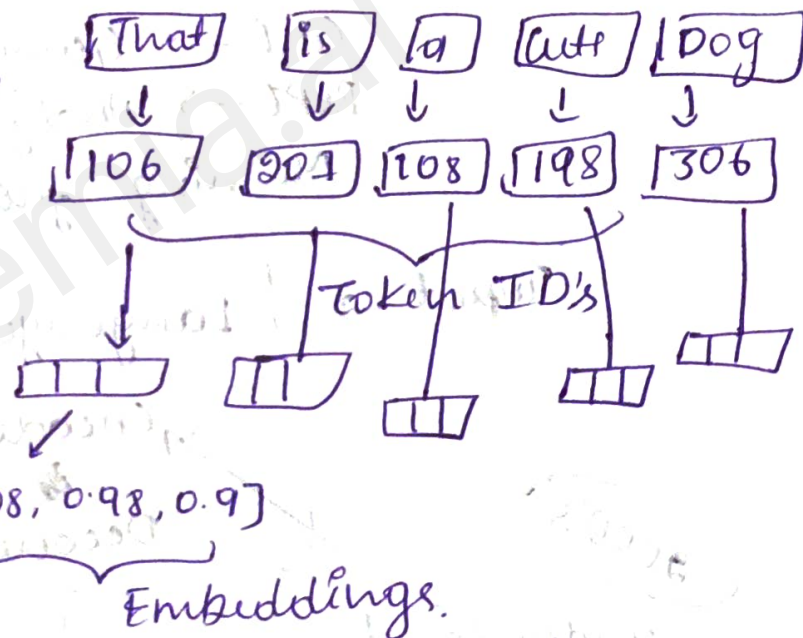
Methods:-

i) Tokenisation



ii) Embeddings

Converting words into
Numbers



(iii) Transformer / Self Attention:-

To predict the pronouns.

ex:-

The dog chases the squirrel because
it is having the.....

Here it $\rightarrow$ Squirrel.

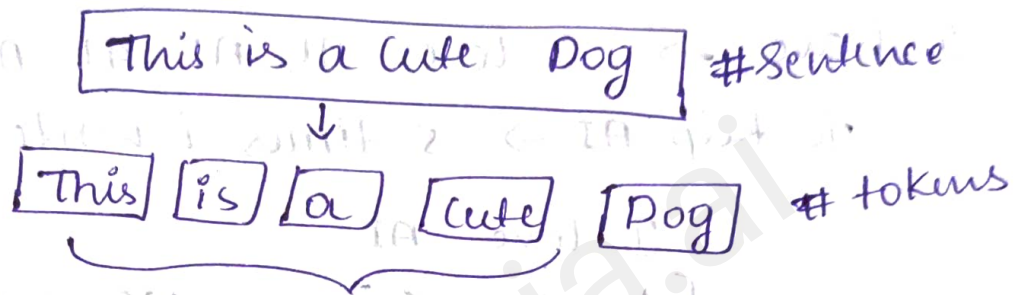
Self attention

[Understand the sentence]

Bag of Words:

Language AI starts with History of Bag of Words
~ 2000's

Suppose our first step we want to create numerical Representation so we need to split like tokenization. as we seen earlier, to create a Numerical Representation.



Method of split is whitespace b/w words.

Suppose if there is no whitespace
e.g. man-darin, this makes incorrect in bag of words [Limitation]

- Also it ignores word order & Semantic Representation [Limit]

Working Strategy :-

[I love AI and I love python]

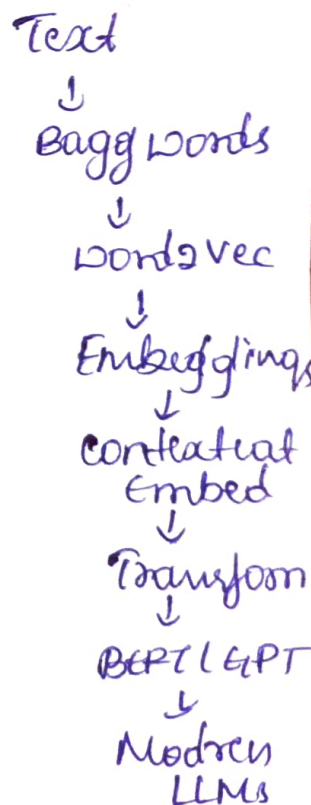
→ Finds unique words.

[I , love , AI , and Python]

have meaning

I → 2 times

love → 2 times.



word I love AI and Python

count 2 2 1 1 1

so the vector becomes:

$[2, 2, 1, 1, 1]$



Imagine if

I have $\rightarrow$ I love AI, AI, AI, AI, AI

To keep AI $\rightarrow$ 5 times i write

I love AI

$[1, 1, 5]$

$\rightarrow$ we need numerical representation

This has limit due to word order &

no meaning of semantic. so we moved to "Word2Vec"

Check the History

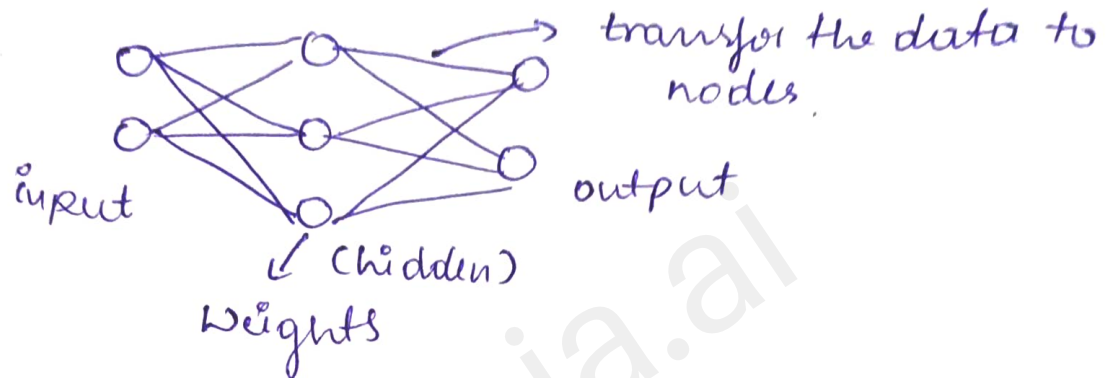
Bag of word
~2000

Word2Vec
2013

Better representation with Dense Vector Embedding:

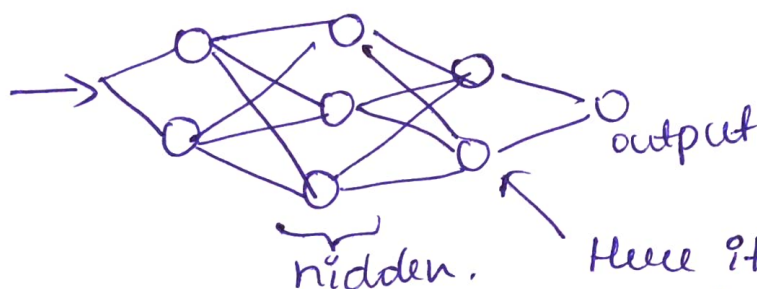
2013, Word2Vec one of the first attempt at capturing the meaning of text in embeddings.

- This uses neural Network to process information.
- In Neural Network each nodes consists of a weight depending on the input is called parameters.



Input $\rightarrow$ I drink coffee $\rightarrow [0.72, 0.42, 0.56 \dots]$ upto 50 we can initialize random value.

$\swarrow$
neural Network to understand the meaning



Here it will check neighbour words

drink $\Leftrightarrow$ coffee

random $\rightarrow [0.42 \quad 0.98]$

updated after training $\rightarrow [0.64 \quad 0.72]$

Near values.